ASSIGNMENT-19.4

HT.NO-2403A52L06

BATCH-50

QUESTION:

Lab 19: Assignment – Code Translation: Converting Between Programming Languages Week 10

Lab Objectives

• Translate code between programming languages while preserving logic and structure.

• Identify language-specific differences in syntax, typing, and execution models.

• Validate AI-generated translations for correctness and performance.

• Compare procedural, object-oriented, and database query translations across ecosystems.

Learning Outcome

After completing this assignment, students will be able to:

• Integrate external REST APIs using AI-assisted Python scripting.

• Implement structured error handling for API failures and invalid inputs.

• Safely parse and validate JSON responses from external services.

• Develop resilient applications capable of handling real-world API limitations.

Task 1 – JavaScript to Python (Array Processing Logic)

Scenario

A frontend application written in JavaScript processes user scores. The backend team wants the same logic implemented in Python for analytics processing.

Task

Translate a JavaScript function that:

• Takes an array of numbers

• Filters values greater than 50

• Returns the average of filtered values

Instructions

• Prompt AI to convert JavaScript array methods (filter, reduce) into equivalent Python logic.

• Ensure list comprehension or Pythonic style is used.

• Compare outputs of both implementations with sample input.

• Document differences in handling arrays between JS and Python.

Expected Output

• Equivalent Python function.

• Same output for identical inputs.

• Clean and Pythonic implementation.

Task 2 – Python OOP to C++ Class Translation

Scenario

A data modeling module is written in Python using OOP. The system is being migrated to a high-performance C++ environment.

Task

Translate a Python class that:

• Contains private attributes

• Uses getter/setter methods

• Includes a method that calculates a derived value

Instructions

• Convert the Python class into a C++ class.

• Ensure:

o Proper access modifiers

o Constructor handling

o Correct data types

• Compile and test the C++ version.

• Compare how encapsulation differs between Python and C++.

Expected Output

• Fully working C++ class equivalent to Python version.

• Proper use of access specifiers (private, public).

Task 3 – REST API Call Translation: Python to JavaScript

Scenario

A backend developer wrote an API request using Python requests. The frontend team needs the same functionality in JavaScript (Fetch API).

Task

Translate Python API request code into JavaScript.

Instructions

• Convert:

o GET request logic

o JSON parsing

o Error handling (try-except → try-catch)

• Ensure proper async/await usage in JavaScript.

• Compare differences in error handling mechanisms.

Expected Output

• Equivalent JavaScript code using Fetch API.

• Proper async handling.

• Similar output structure.

Task 4 – SQL Aggregation to MongoDB Query Translation

Scenario

A company is migrating from relational databases (SQL) to MongoDB (NoSQL).

Task

Translate an SQL aggregation query into a MongoDB aggregation pipeline.

Example Logic:

• Group records by category

• Calculate average price

• Filter groups where average price > threshold

Instructions

• Provide SQL query to AI.

• Generate equivalent MongoDB query.

• Explain differences between relational and document-based querying.

• Validate results conceptually.

Expected Output

• Working MongoDB aggregation query.

• Clear understanding of grouping and aggregation differences.

Task 5 – Real-Time Application: Algorithm Translation Across Three Languages

Scenario

A tech company maintains codebases in multiple languages (Python, Java, and Go). A common algorithm must be implemented consistently across systems.

Task

Choose an algorithm (e.g., binary search, palindrome check, prime number checker) and translate it across:

• Python

• Java

• One additional language (Go / C# / JavaScript)

Instructions

• Ensure logic remains identical across languages.

• Handle:

o Data types

o Loop differences

o Input/output handling

• Test each version with same input values.

• Briefly document translation challenges.

Expected Output

• Three equivalent working implementations.

• Clear logical consistency.

• Understanding of syntactic vs structural differences.

CODE:

```
import json
from typing import Dict, Any, Optional


# ==========================================================================
# TASK 1: JavaScript to Python (Array Processing Logic)
# ==========================================================================


def filter_and_average(scores):
    """
    Translates JavaScript array processing logic to Python.

    JavaScript equivalent:
    function filterAndAverage(scores) {
        return scores
            .filter(val => val > 50)
            .reduce((sum, val) => sum + val, 0) / scores.filter(val => val > 50).length;
    }

    Args:
```

```
        scores: List of numbers

    Returns:
        Average of values greater than 50, or 0 if none exist
    """
    filtered = [score for score in scores if score > 50]

    if len(filtered) == 0:
        return 0

    return sum(filtered) / len(filtered)

# Test Task 1
print("=" * 70)
print("TASK 1: JavaScript to Python (Array Processing)")
print("=" * 70)
test_scores = [30, 45, 55, 60, 70, 40, 80, 50]
result_task1 = filter_and_average(test_scores)
print(f"Input: {test_scores}")
print(f"Filtered values (>50): {[x for x in test_scores if x > 50]}")
print(f"Average of filtered values: {result_task1}")
print()

# ==========================================================================
# TASK 2: Python OOP to C++ Class Translation
# ==========================================================================
```

```
class Student:
    """
    Python class demonstrating OOP concepts.
    Will be translated to C++ with equivalent functionality.

    Attributes:
        __name (private): Student name
        __gpa (private): Grade Point Average
        __credits (private): Course credits
    """

    def __init__(self, name, gpa, credits):
        """Constructor"""
        self.__name = name
        self.__gpa = gpa
        self.__credits = credits

    def get_name(self):
        """Getter for name"""
        return self.__name

    def set_name(self, name):
        """Setter for name"""
        self.__name = name

    def get_gpa(self):
        """Getter for GPA"""
        return self.__gpa
```

```
def set_gpa(self, gpa):
    """Setter for GPA"""
    if 0 <= gpa <= 4.0:
        self.__gpa = gpa
    else:
        raise ValueError("GPA must be between 0 and 4.0")

def get_credits(self):
    """Getter for credits"""
    return self.__credits

def set_credits(self, credits):
    """Setter for credits"""
    self.__credits = credits

def calculate_academic_standing(self):
    """
    Calculates derived value: academic standing.
    Returns: "Excellent" if GPA >= 3.5, "Good" if >= 3.0, "Fair" otherwise
    """
    if self.__gpa >= 3.5:
        return "Excellent"
    elif self.__gpa >= 3.0:
        return "Good"
    else:
        return "Fair"

def __str__(self):
```

```
        return f"Student: {self.__name}, GPA: {self.__gpa}, Credits: {self.__credits}"

# Test Task 2
print("=" * 70)
print("TASK 2: Python OOP Class")
print("=" * 70)
student1 = Student("Alice Johnson", 3.8, 120)
print(f"Created: {student1}")
print(f"Academic Standing: {student1.calculate_academic_standing()}")
student1.set_gpa(3.2)
print(f"Updated GPA: {student1.get_gpa()}")
print(f"New Academic Standing: {student1.calculate_academic_standing()}")
print()

# ==========================================================================
# TASK 3: REST API Call Translation (Python to JavaScript concept)
# ==========================================================================

def fetch_user_data_python(user_id: int) -> Optional[Dict[str, Any]]:
    """
    Python implementation using requests library for API calls.

    JavaScript Equivalent (Fetch API):
    async function fetchUserData(userId) {
        try {
            const response = await fetch(`https://api.example.com/users/${userId}`);
```

```
        if (!response.ok) {
            throw new Error(`HTTP error! status: ${response.status}`);
        }
        const data = await response.json();
        return data;
    } catch (error) {
        console.error('Error:', error.message);
        return null;
    }
}
"""
try:
    # Note: Using mock data instead of actual API call for testing
    if user_id <= 0:
        raise ValueError("User ID must be positive")

    # Simulated API response
    mock_response = {
        "id": user_id,
        "name": "John Doe",
        "email": "john@example.com",
        "status": "active"
    }

    return mock_response

except ValueError as e:
    print(f"Error: {str(e)}")
    return None
```

```
    except Exception as e:
        print(f"Unexpected error: {str(e)}")
        return None

# Test Task 3
print("=" * 70)
print("TASK 3: REST API Call - Python Implementation")
print("=" * 70)
result_task3 = fetch_user_data_python(1)
if result_task3:
    print(f"API Response: {json.dumps(result_task3, indent=2)}")
print()

# ======================================================================
# TASK 4: SQL to MongoDB Query Translation
# ======================================================================

def sql_to_mongodb_explanation():
    """
    SQL Query:
    SELECT category, AVG(price) as average_price
    FROM products
    GROUP BY category
    HAVING AVG(price) > 50
    ORDER BY average_price DESC;
```

MongoDB Equivalent (Aggregation Pipeline):

```
db.products.aggregate([
    {
        $group: {
            _id: "$category",
            average_price: { $avg: "$price" }
        }
    },
    {
        $match: { average_price: { $gt: 50 } }
    },
    {
        $sort: { average_price: -1 }
    }
])
"""
return {
    "SQL_Query": """
        SELECT category, AVG(price) as average_price
        FROM products
        GROUP BY category
        HAVING AVG(price) > 50
        ORDER BY average_price DESC;
    """,
    "MongoDB_Pipeline": [
        {"$group": {"_id": "$category", "average_price": {"$avg": "$price"}}},
        {"$match": {"average_price": {"$gt": 50}}},
        {"$sort": {"average_price": -1}}
    ],
```

```
        "Differences": {
            "Grouping": "SQL uses GROUP BY; MongoDB uses $group stage",
            "Filtering": "SQL uses HAVING clause; MongoDB uses $match stage",
            "Sorting": "Both use ORDER BY / $sort, but MongoDB uses -1 for DESC",
            "Document_Structure": "MongoDB returns documents; SQL returns tuples",
            "Flexibility": "MongoDB is more flexible with nested documents"
        }
    }

# Test Task 4
print("=" * 70)
print("TASK 4: SQL to MongoDB Query Translation")
print("=" * 70)
task4_result = sql_to_mongodb_explanation()
print("SQL Query:")
print(task4_result["SQL_Query"].strip())
print("\nMongoDB Aggregation Pipeline:")
print(json.dumps(task4_result["MongoDB_Pipeline"], indent=2))
print("\nKey Differences:")
for key, value in task4_result["Differences"].items():
    print(f"  • {key}: {value}")
print()

# ======================================================================
# TASK 5: Algorithm Translation Across Three Languages
# ======================================================================
```

```python
def binary_search_python(arr: list, target: int) -> int:
    """
    Python implementation of binary search algorithm.

    Args:
        arr: Sorted list of integers
        target: Value to search for

    Returns:
        Index of target if found, -1 otherwise
    """
    left, right = 0, len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return -1

def binary_search_java_code():
    """Returns Java equivalent of binary search"""
    return """
```

```
    public class BinarySearch {
        public static int binarySearch(int[] arr, int target) {
            int left = 0, right = arr.length - 1;

            while (left <= right) {
                int mid = (left + right) / 2;

                if (arr[mid] == target) {
                    return mid;
                } else if (arr[mid] < target) {
                    left = mid + 1;
                } else {
                    right = mid - 1;
                }
            }

            return -1;
        }
    }
    """

def binary_search_go_code():
    """Returns Go equivalent of binary search"""
    return """
    package main

    func BinarySearch(arr []int, target int) int {
        left, right := 0, len(arr)-1
```

```
        for left <= right {
            mid := (left + right) / 2

            if arr[mid] == target {
                return mid
            } else if arr[mid] < target {
                left = mid + 1
            } else {
                right = mid - 1
            }
        }

        return -1
    }
    """

# Test Task 5
print("=" * 70)
print("TASK 5: Binary Search Algorithm - Three Language Implementation")
print("=" * 70)

test_array = [2, 5, 8, 12, 16, 23, 38, 45, 56, 67]
test_target = 23

# Python Version
result_python = binary_search_python(test_array, test_target)
print(f"Python Implementation:")
print(f"  Array: {test_array}")
print(f"  Target: {test_target}")
```

```
print(f"  Result: Index {result_python}")
print()

# Java Version Code
print("Java Implementation:")
print(binary_search_java_code())
print()

# Go Version Code
print("Go Implementation:")
print(binary_search_go_code())
print()
```

OUTPUT:

```
PS C:\Users\akhir\OneDrive\Documents\AI ASSISTANT> & C:\Users\akhir\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/akhir/OneDrive/Documents/AI ASSISTANT/19.4.py"
======================================================================
TASK 1: JavaScript to Python (Array Processing)
======================================================================
Input: [30, 45, 55, 60, 70, 40, 80, 50]
Filtered values (>50): [55, 60, 70, 80]
Average of filtered values: 66.25

======================================================================
TASK 2: Python OOP Class
======================================================================
Created: Student: Alice Johnson, GPA: 3.8, Credits: 120
Academic Standing: Excellent
Updated GPA: 3.2
New Academic Standing: Good

======================================================================
TASK 3: REST API Call - Python Implementation
======================================================================
API Response: {
  "id": 1,
  "name": "John Doe",
  "email": "john@example.com",
  "status": "active"
}
```

```
======================================================================
TASK 4: SQL to MongoDB Query Translation
======================================================================
SQL Query:
SELECT category, AVG(price) as average_price
            FROM products
            GROUP BY category
            HAVING AVG(price) > 50
            ORDER BY average_price DESC;

MongoDB Aggregation Pipeline:
[
  {
    "$group": {
      "_id": "$category",
      "average_price": {
        "$avg": "$price"
      }
    }
  },
  {
    "$match": {
      "average_price": {
        "$gt": 50
      }
    }
  },
  {
    "$sort": {
      "average_price": -1
    }
  }
]

Key Differences:
  • Grouping: SQL uses GROUP BY; MongoDB uses $group stage
  • Filtering: SQL uses HAVING clause; MongoDB uses $match stage
  • Sorting: Both use ORDER BY / $sort, but MongoDB uses -1 for DESC
  • Document_Structure: MongoDB returns documents; SQL returns tuples
  • Flexibility: MongoDB is more flexible with nested documents
```

```
======================================================================
TASK 5: Binary Search Algorithm - Three Language Implementation
======================================================================
Python Implementation:
  Array: [2, 5, 8, 12, 16, 23, 38, 45, 56, 67]
  Target: 23
  Result: Index 5

Java Implementation:

    public class BinarySearch {
        public static int binarySearch(int[] arr, int target) {
            int left = 0, right = arr.length - 1;

            while (left <= right) {
                int mid = (left + right) / 2;

                if (arr[mid] == target) {
                    return mid;
                } else if (arr[mid] < target) {
                    left = mid + 1;
                } else {
                    right = mid - 1;
                }
            }

            return -1;
        }
    }


Go Implementation:

    package main

    func BinarySearch(arr []int, target int) int {
        left, right := 0, len(arr)-1

        for left <= right {
            mid := (left + right) / 2
```

```
        for left <= right {
            mid := (left + right) / 2

            if arr[mid] == target {
                return mid
            } else if arr[mid] < target {
                left = mid + 1
            } else {
                right = mid - 1
            }
        }

        return -1
    }


PS C:\Users\akhir\OneDrive\Documents\AI ASSISTANT>
```